ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO TECHNICAL PROGRAMMING

PAPER 2 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Paper 2 (solution development). Section A: objective questions (30) · Section B: C++ input and output with cin and cout, selection and switch, loops, functions and variable scope, arrays, an introduction to classes and objects, and block-based Arduino programming (70).

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INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write all program code in C++ unless the question says otherwise. Small spelling mistakes in code will not be penalised if the logic is correct.
5. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 In C++, cout is used to …

A read input from the keyboard.
B declare a constant.
C end the program.
D display output.

1.1.2 The operator used with cin is …

A <<
B >>
C ==
D ++

1.1.3 Which statement selects one of many cases using the value of a variable?

A for
B while
C return
D switch

1.1.4 In a switch statement, the keyword that stops execution falling into the next case is …

A stop
B break
C exit
D continue

1.1.5 A variable declared inside a function can only be used inside that function. This is called … scope.

A global
B static
C local
D public

1.1.6 A function that does not return a value has the return type …

A void
B int
C double
D bool

1.1.7 The index of the first element of a C++ array is …

A 1
B 0
C −1
D the array size

1.1.8 In object-oriented programming, the variables that belong to a class are called …

A methods.
B libraries.
C parameters only.
D attributes.

1.1.9 An object is …

A an instance of a class.
B a type of loop.
C a compiler error.
D a library file.

1.1.10 mBlock is an example of a …

A text editor for C. B 3D printer slicer.
C block-based programming environment. D database program.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 C++ is case sensitive, so Total and total are different variables.

1.2.2 The statement using namespace std; lets you write cout instead of std::cout.

1.2.3 A for loop is best when you know how many times to repeat.

1.2.4 A do…while loop may not run at all.

1.2.5 Parameters are values passed into a function.

1.2.6 A function can return more than one value with a single return statement.

1.2.7 An array can store values of different data types.

1.2.8 Reading array element [5] of an array declared as int a[5] is outside the array.

1.2.9 A class is a blueprint from which objects are created.

1.2.10 Methods are functions that belong to a class.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Header file needed for cin and cout	A	iostream
1.3.2	Operator that means "and" in a condition	B	||
1.3.3	Operator that means "or" in a condition	C	&&
1.3.4	Adds 1 to a variable	D	default
1.3.5	Case that runs when no other case matches	E	function prototype
1.3.6	Variable declared outside all functions	F	global variable
1.3.7	Declaring a function before main() so it can be called	G	constructor
1.3.8	Special method that runs when an object is created	H	++
1.3.9	Hiding the data of a class and allowing access only through methods	I	encapsulation
1.3.10	Moving through a program line by line to find errors	J	stepping (debugger)

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: INPUT, OUTPUT AND SELECTION

2.1 Write a C++ program that reads a temperature in °C from a robot's sensor (typed by the user) and displays it in °F using F = C × 9 ÷ 5 + 32. Use cin and cout. (6)

2.2 A robot receives a single-character command: 'F' forward, 'B' back, 'L' left, 'R' right. Write a switch statement that displays the action for each command and "Unknown command" for any other character. (7)

2.3 Explain why it is better to use a switch statement than many if-statements for the menu in 2.2. (2)

[15]

QUESTION 3: LOOPS

3.1 Study the code below and answer the questions.

```
int n = 1;
int sum = 0;
do {
    sum = sum + n * n;
    n = n + 2;
} while (n <= 7);
cout << sum;
```

3.1.1 Draw a trace table for n and sum. (5)

3.1.2 What is displayed? (1)

3.2 Write a loop that asks the user for battery readings until the user enters −1, and then displays the number of readings and the average reading. (8)

[14]

QUESTION 4: FUNCTIONS AND ARRAYS

4.1 Write a function named area that takes the length and width of a rectangular robot base (in cm) as parameters and returns the area. Show how it is called from main(). (6)

4.2 Explain the difference between a local and a global variable, and say why global variables should be used carefully. (3)

4.3 The distances (in cm) measured by a robot's ultrasonic sensor are stored in an array:

```
int dist[6] = {45, 12, 80, 33, 9, 60};
```

4.3.1 What is the value of dist[3]? (1)

4.3.2 Write code that finds and displays the smallest distance in the array. (6)

4.3.3 Write code that counts how many distances are less than 20 cm. (4)

[20]

QUESTION 5: CLASSES AND BLOCK PROGRAMMING

5.1 Study the class below and answer the questions.

```
class Robot {
  private:
    string name;
    int battery;          // percentage
  public:
    Robot(string n) { name = n; battery = 100; }
    void drive(int minutes) { battery = battery - minutes * 2; }
    int getBattery() { return battery; }
};
```

5.1.1 Name the attributes and the methods of the class. (3)

5.1.2 Write the code to create a Robot object called rover named "Rover1", let it drive for 15 minutes and display its battery level. (4)

5.1.3 What will the battery level be? (1)

5.1.4 Why are the attributes declared private? (1)

5.2 In mBlock (or another block-based Arduino tool), describe the blocks needed to make an LED on pin 13 blink once per second forever. (3)

[12]

QUESTION 6: DEBUGGING

6.1 The program below should display the numbers 1 to 10, but it contains THREE errors. Identify each error and correct it. (6)

```
#include <iostream>
using namespace std

int main() {
    for (int i = 1; i < 10; i++)
        cout << I << endl;
    return 0;
}
```

6.2 Name THREE features of a debugger that help a programmer find logic errors. (3)

[9]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER